

SUBJECT : PROBLEM SOLVING AND TESTING USING JAVA

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Task 3 (Easy): Maximum Profit Analyzer

Problem Description

Given daily profit/loss values, find the maximum possible profit obtainable from a contiguous sequence of days using Kadane's Algorithm.

Input Format

- First line contains integer N.
- Second line contains N integers.

Output Format

Print maximum subarray sum.

Constraints

- $1 \leq N \leq 10^5$

Sample Input

8
-2 -3 4 -1 -2 1 5 -3

Sample Output

7

PROGRAM :

```
import java.util.*;
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int n = sc.nextInt();  
        int[] arr = new int[n];  
  
        for (int i = 0; i < n; i++) {  
            arr[i] = sc.nextInt();  
        }  
  
        int maxSum = arr[0];  
        int currentSum = arr[0];  
  
        for (int i = 1; i < n; i++) {  
            currentSum = Math.max(arr[i], currentSum + arr[i]);  
            maxSum = Math.max(maxSum, currentSum);  
        }  
  
        System.out.println(maxSum);  
    }  
}
```

OUTPUT :

A terminal window with a dark background and a light gray title bar. The title bar contains a dropdown arrow, a magnifying glass, a document icon, a gear icon, and a trash icon, followed by the text "input". The terminal content shows the number "8" on the first line, the sequence "1 2 -5 -6 4 3 4 2" on the second line, and the number "13" on the third line. Below these, a green message states "...Program finished with exit code 0". At the bottom, a green prompt "js" is followed by the text "Press ENTER to exit console.".

```
8
1 2 -5 -6 4 3 4 2
13

...Program finished with exit code 0
js Press ENTER to exit console.
```